

SPS Sintering Regression Analysis Report

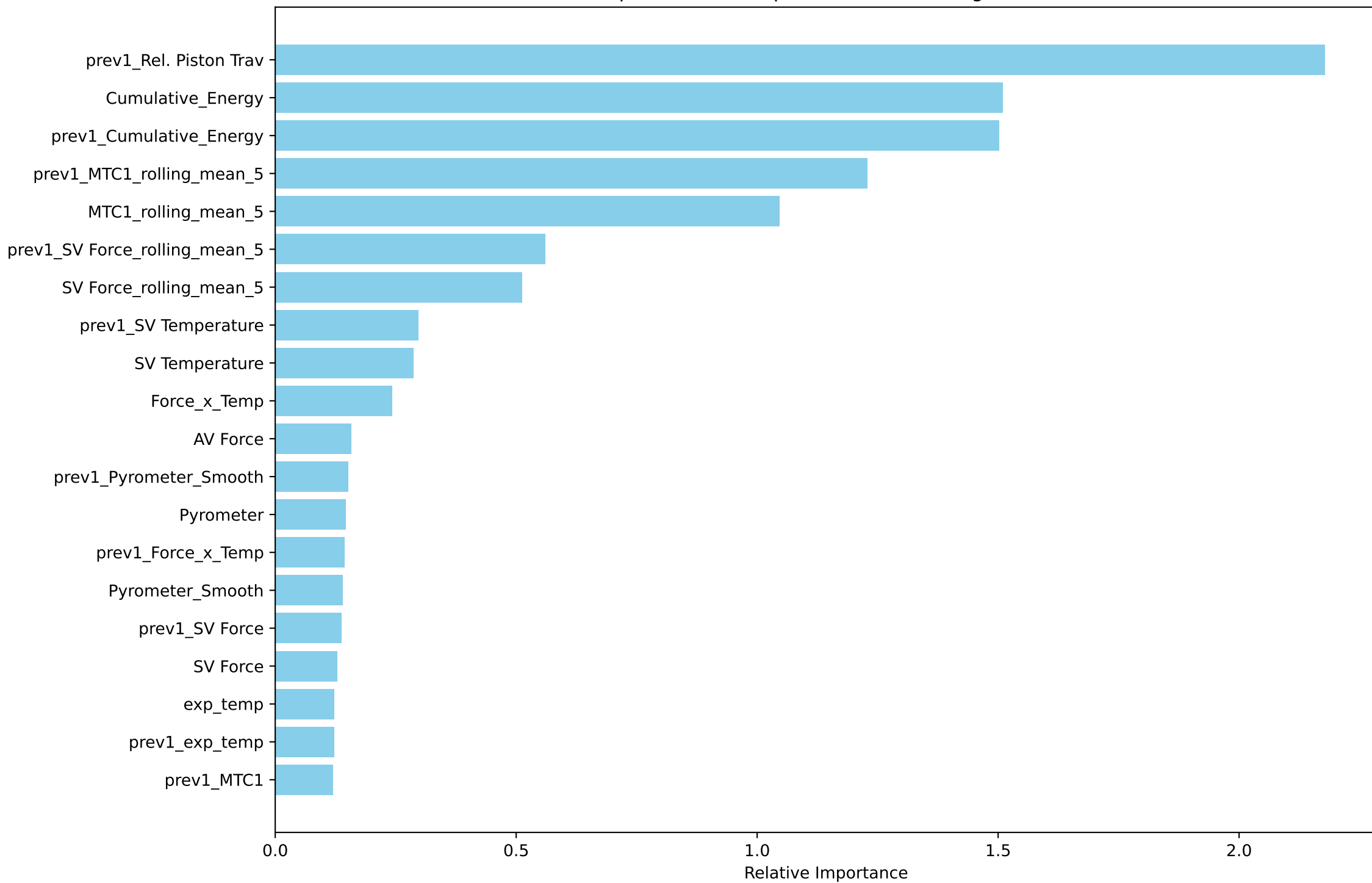
Date: 2026-02-28 01:29

Model Performance (Validation Data):

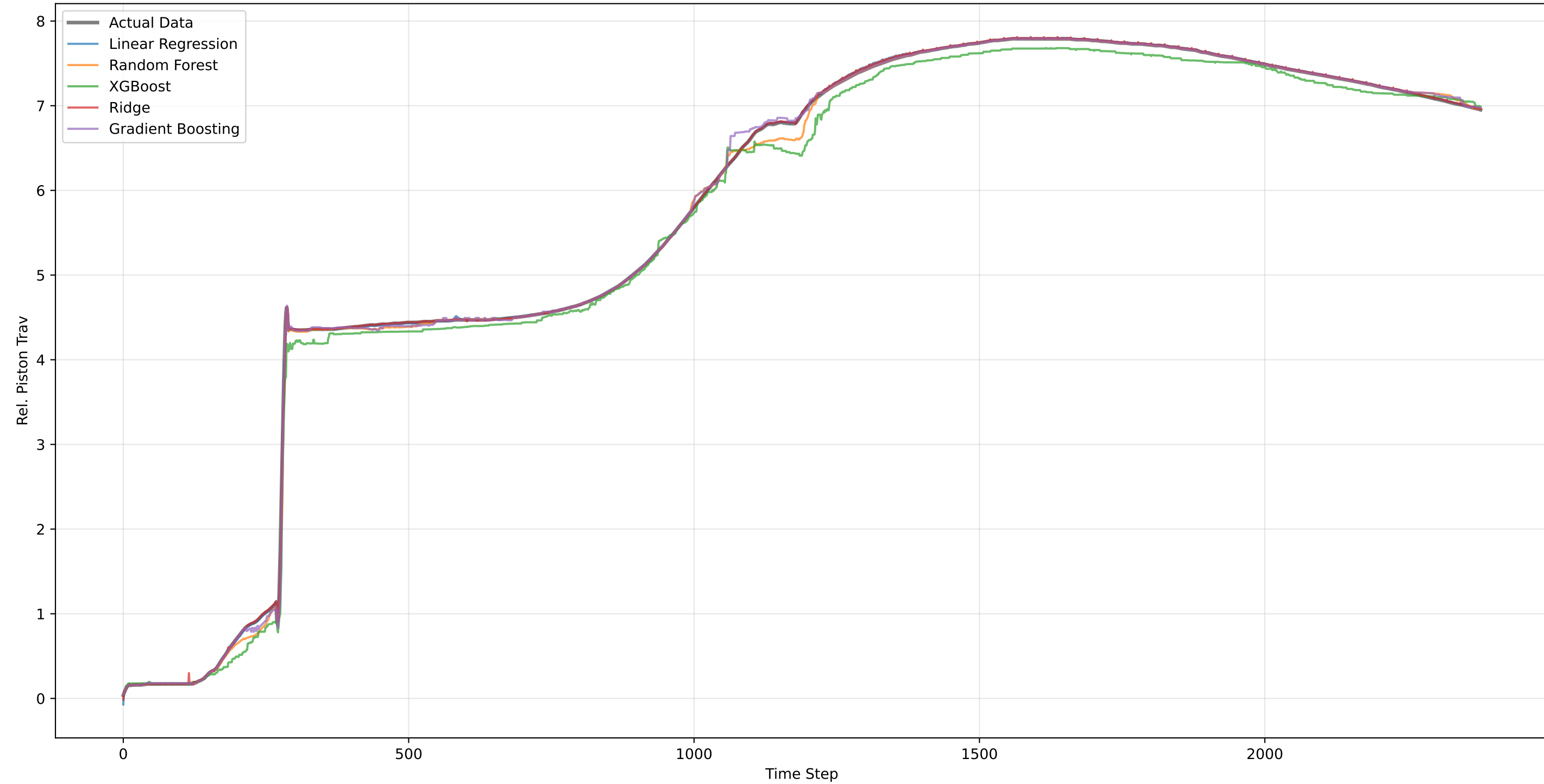
Model	R2	RMSE	MAE
Linear Regression	1.0000	0.0139	0.0046
Random Forest	0.9999	0.0169	0.0031
XGBoost	0.9999	0.0170	0.0050
Ridge	0.9999	0.0170	0.0050
Gradient Boosting	0.9999	0.0175	0.0032
ElasticNet	0.9999	0.0188	0.0054
Lasso	0.9999	0.0197	0.0052
Decision Tree	0.9999	0.0248	0.0045
GPR	0.9998	0.0337	0.0047
KNN	0.9997	0.0379	0.0088
MLP	0.9982	0.0922	0.0582

Best Model Selected: Linear Regression

Top 20 Feature Importance - Linear Regression

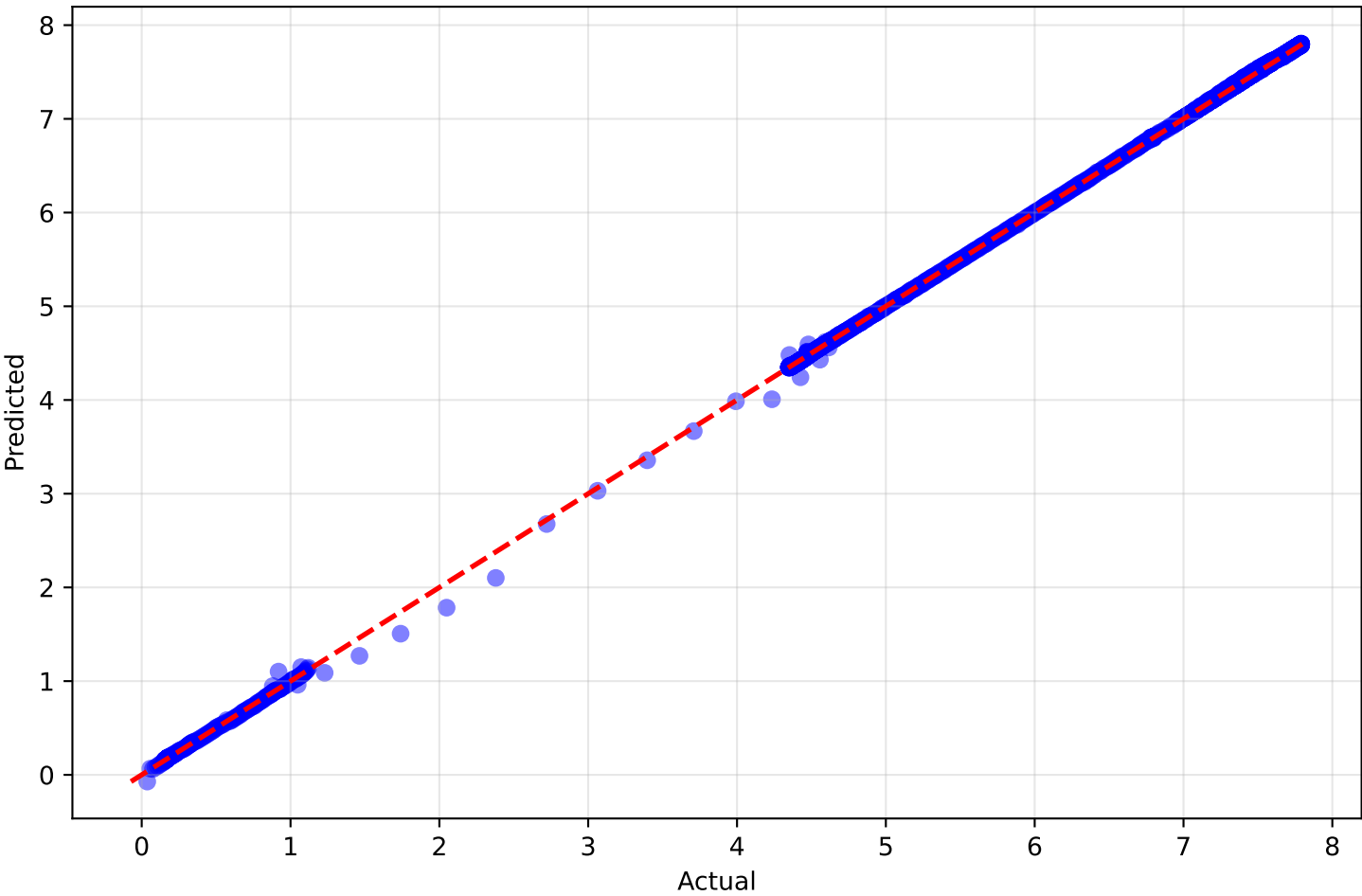


Top 5 Models Comparison (Time Series)

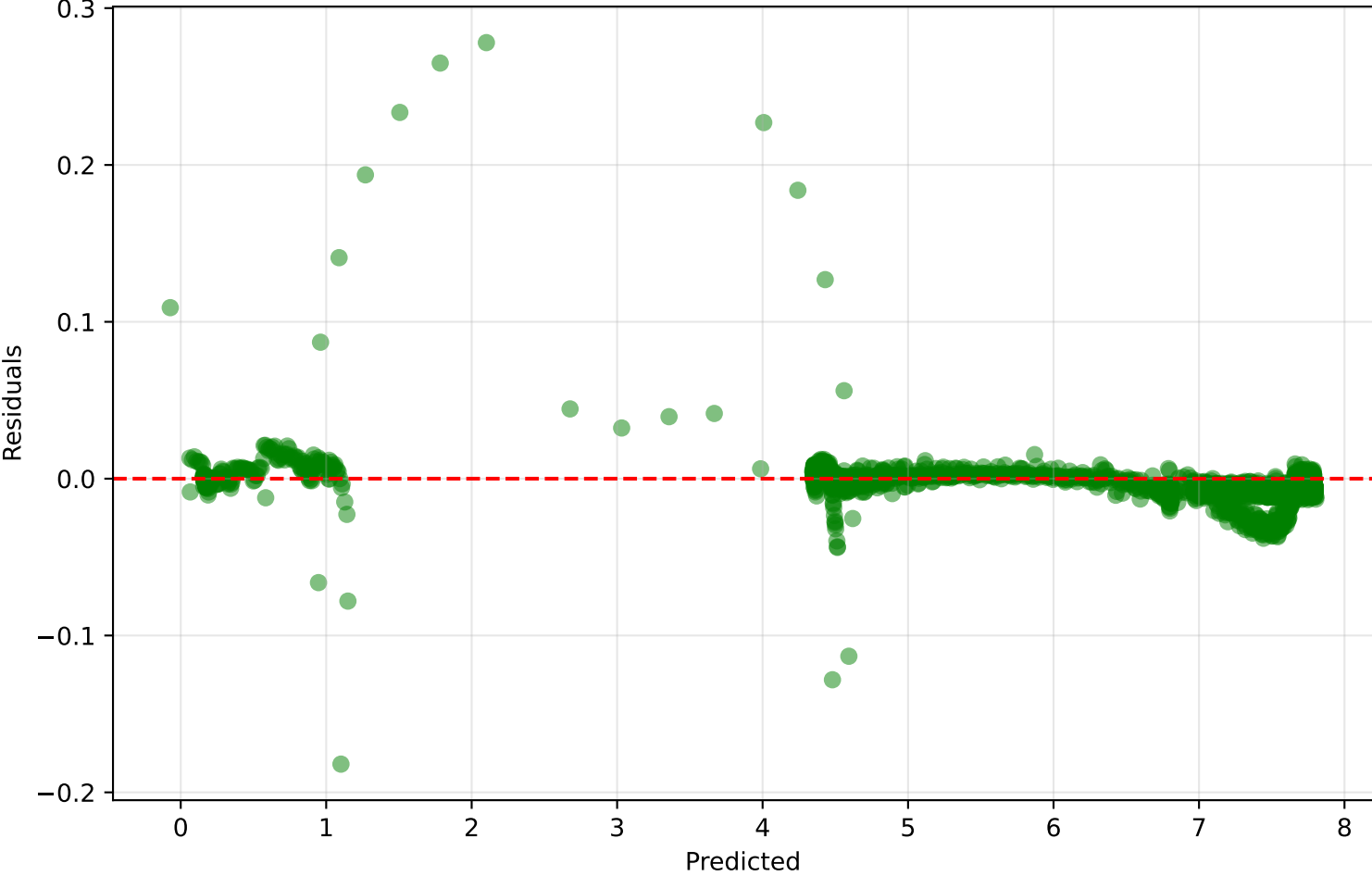


Detailed Analysis: Linear Regression

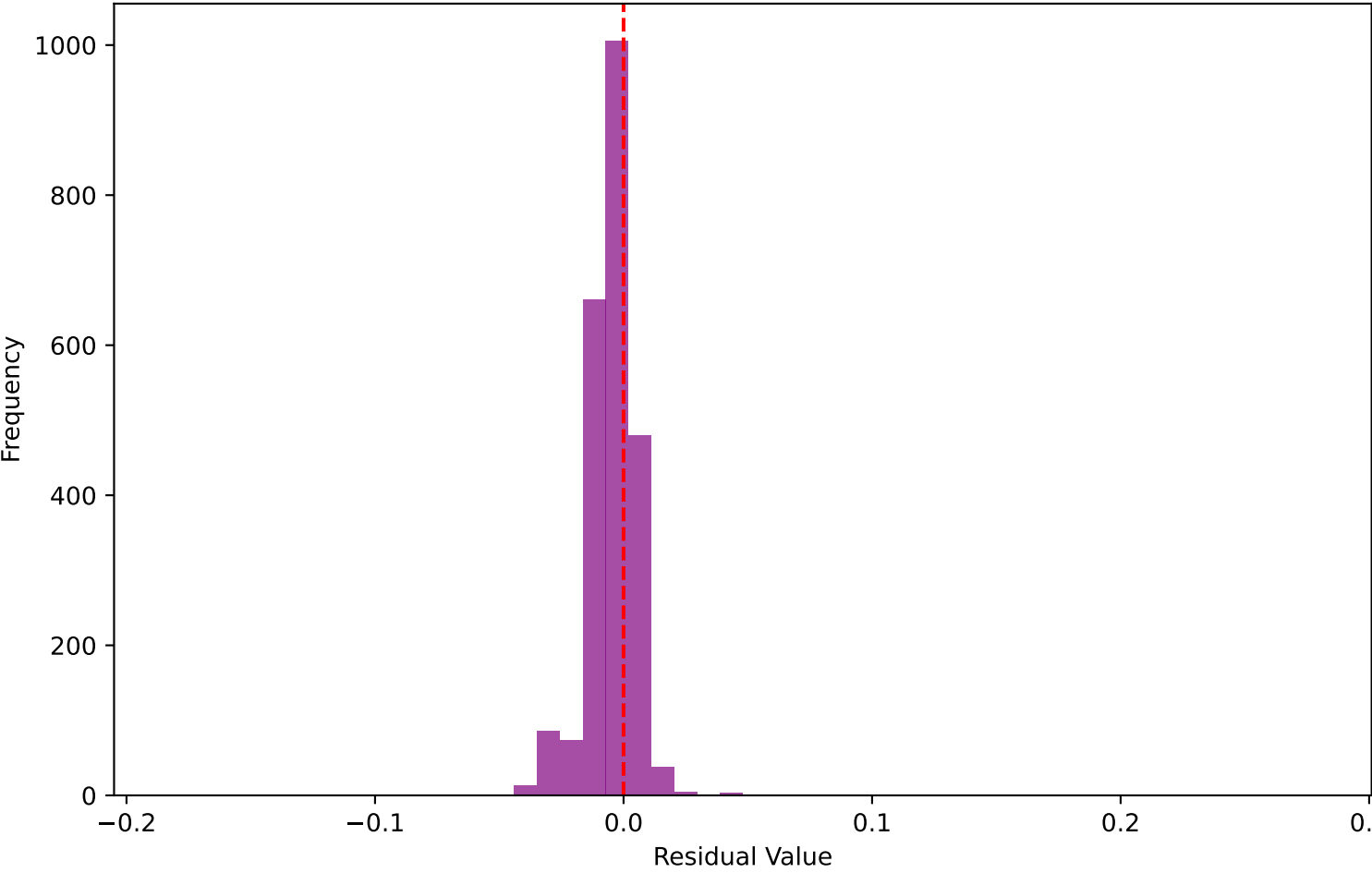
Actual vs Predicted



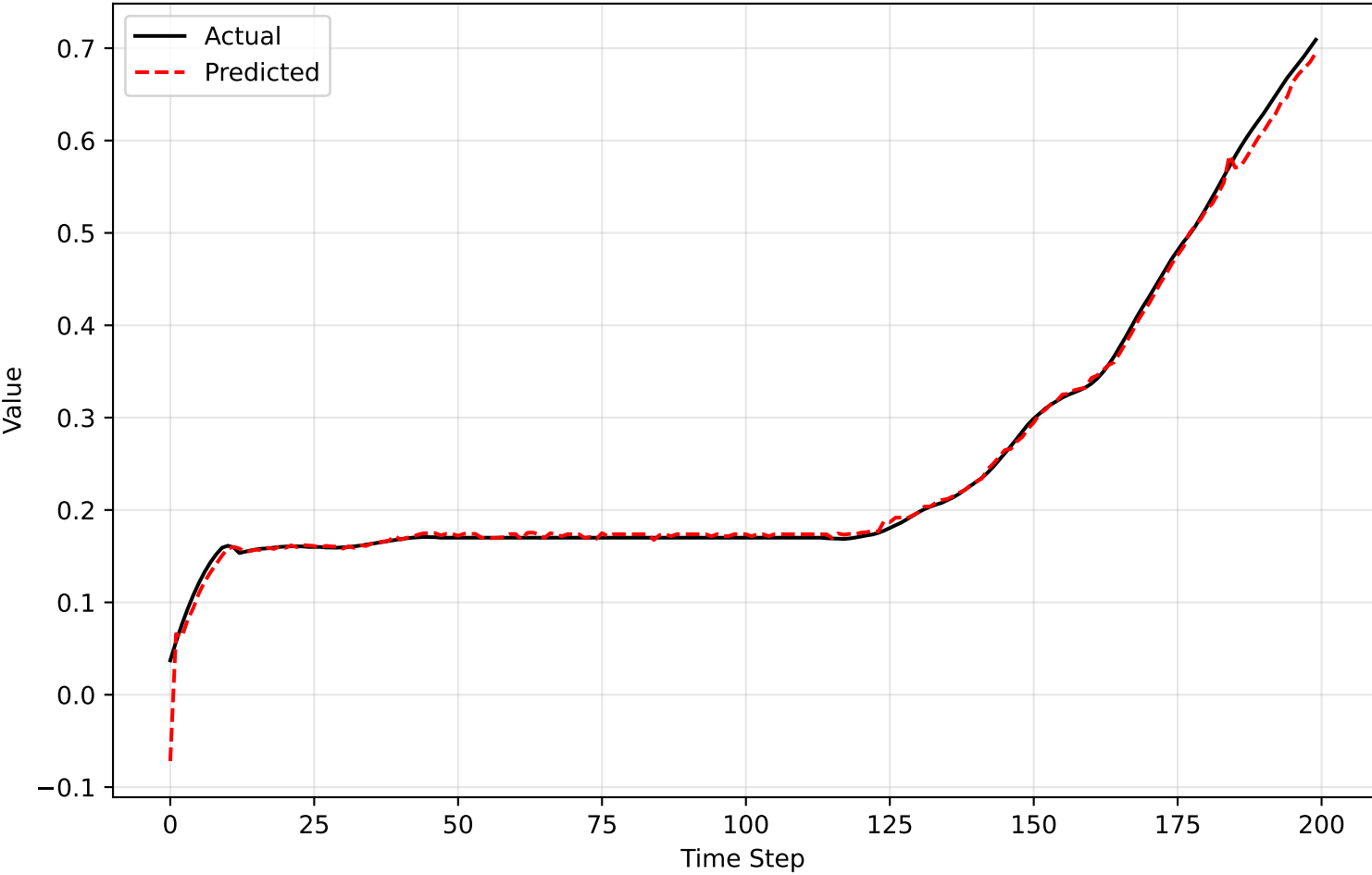
Residuals vs Predicted



Residual Distribution

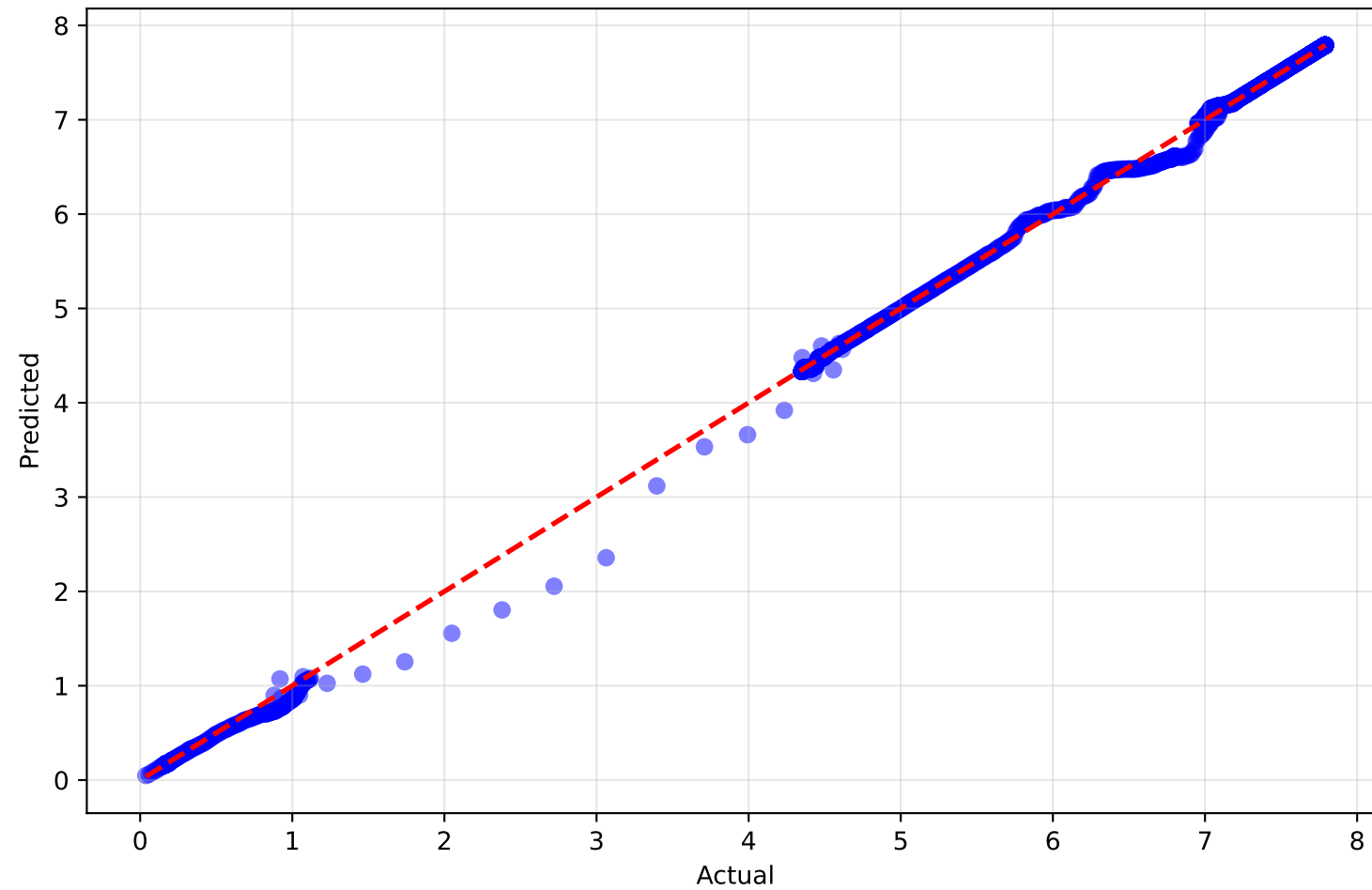


Zoomed View (First 200 points)

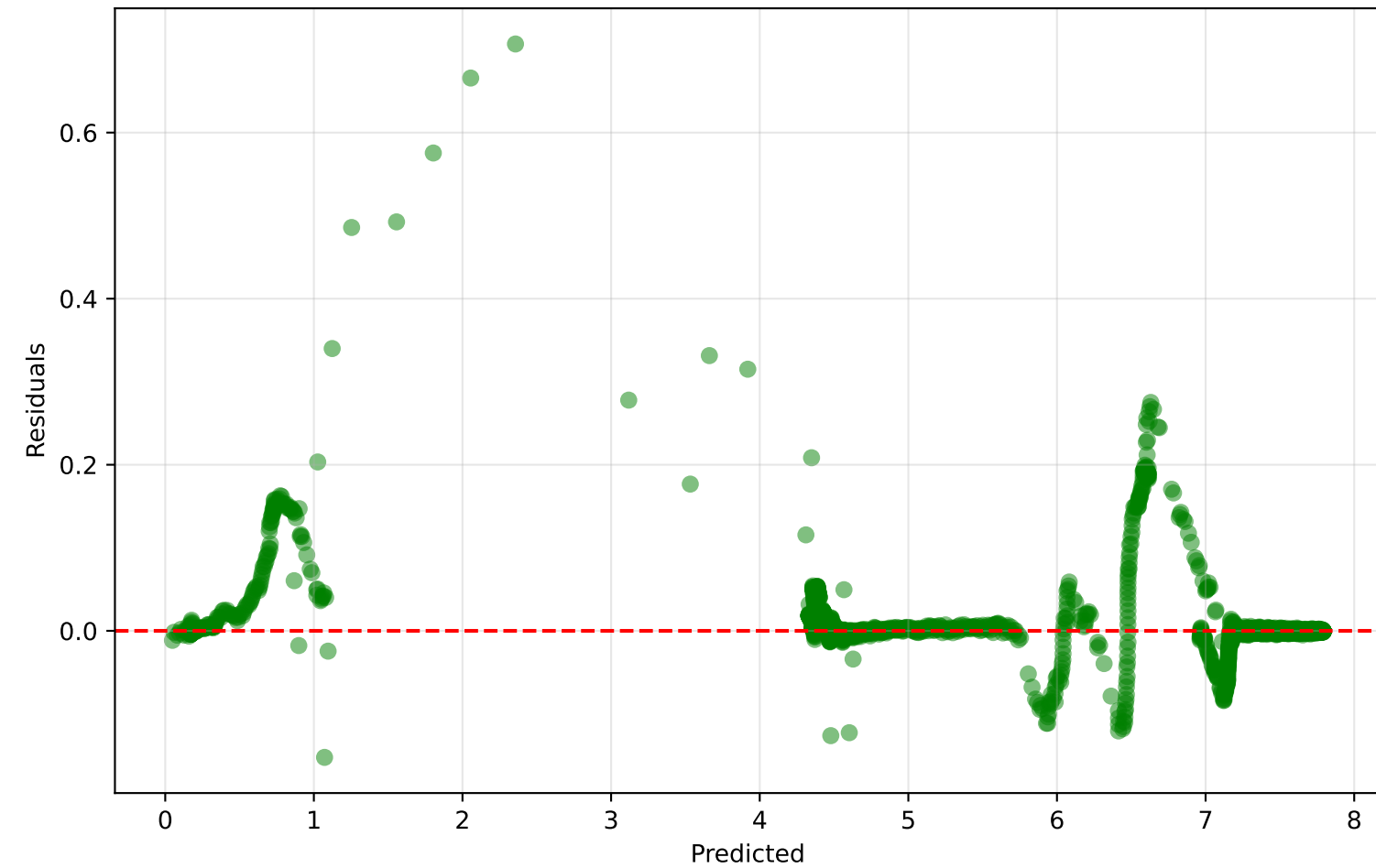


Detailed Analysis: Random Forest

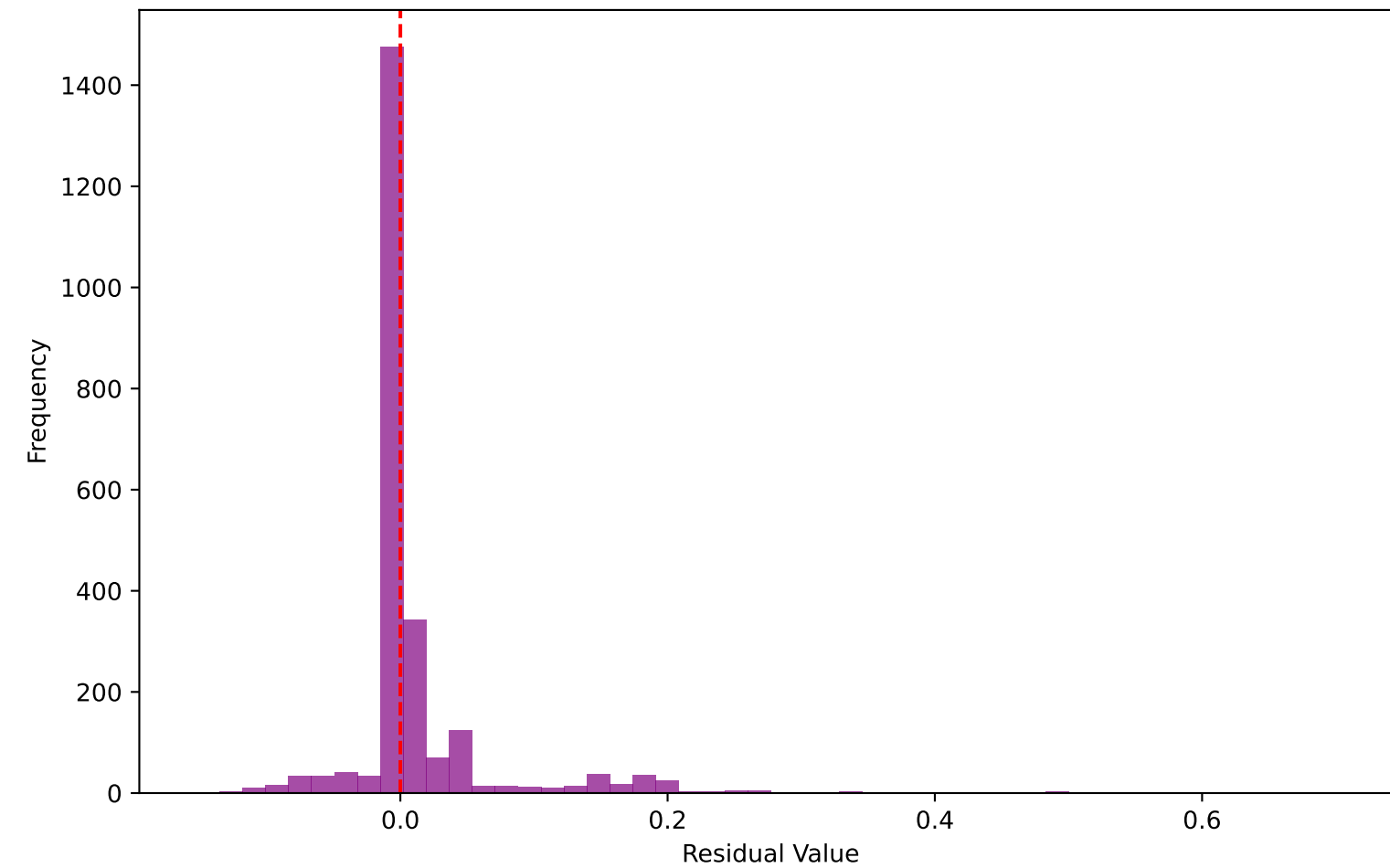
Actual vs Predicted



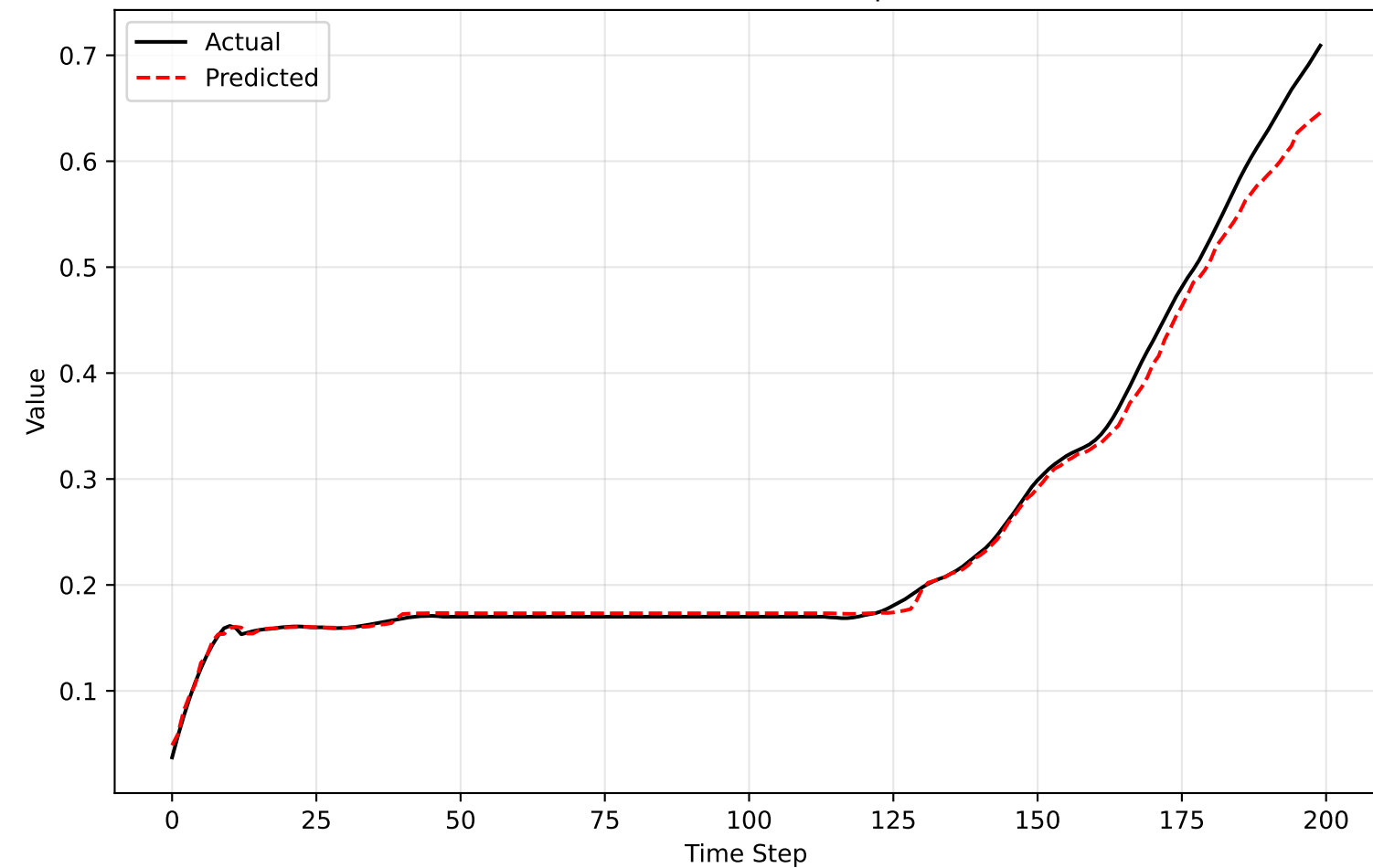
Residuals vs Predicted



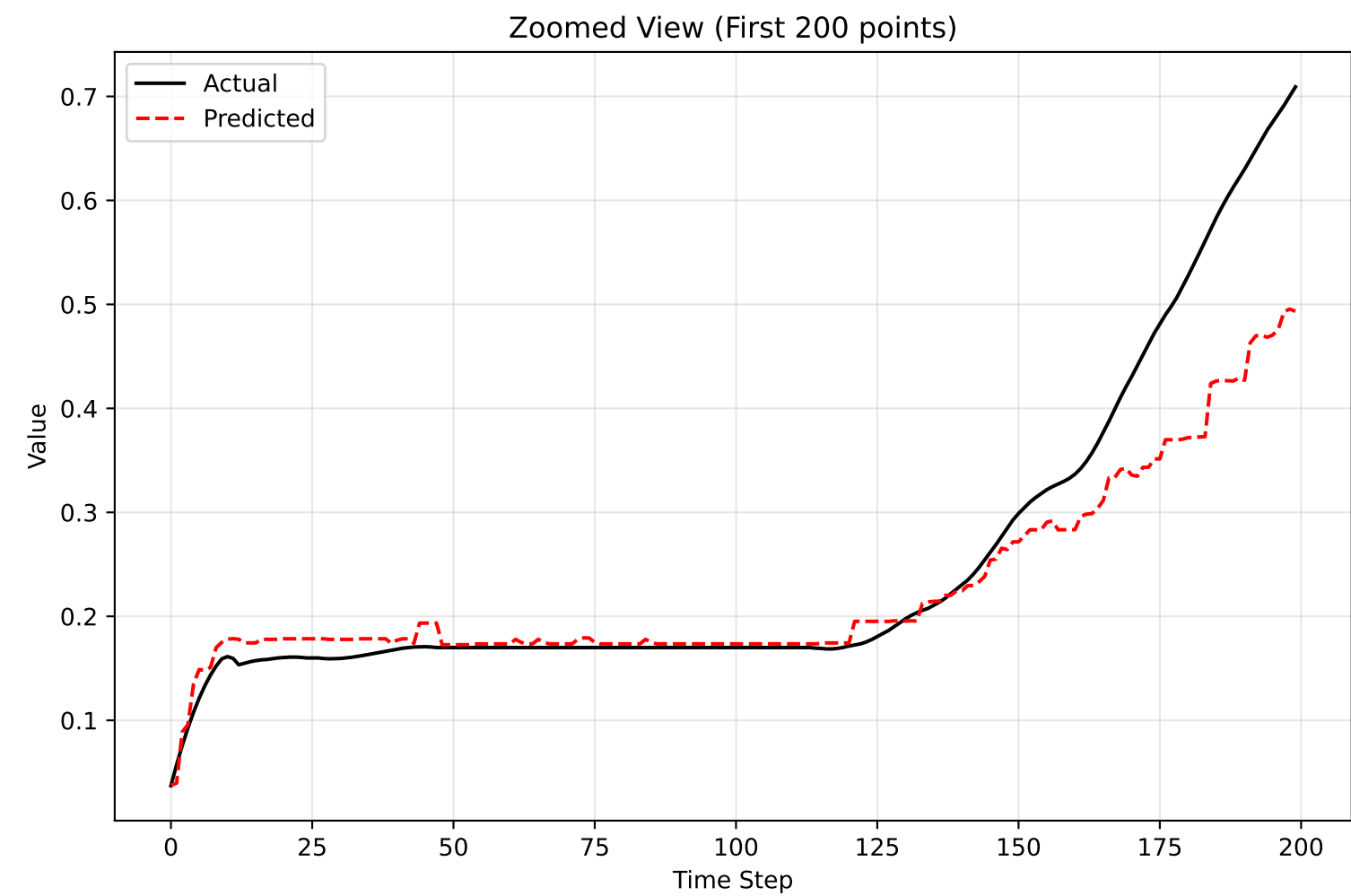
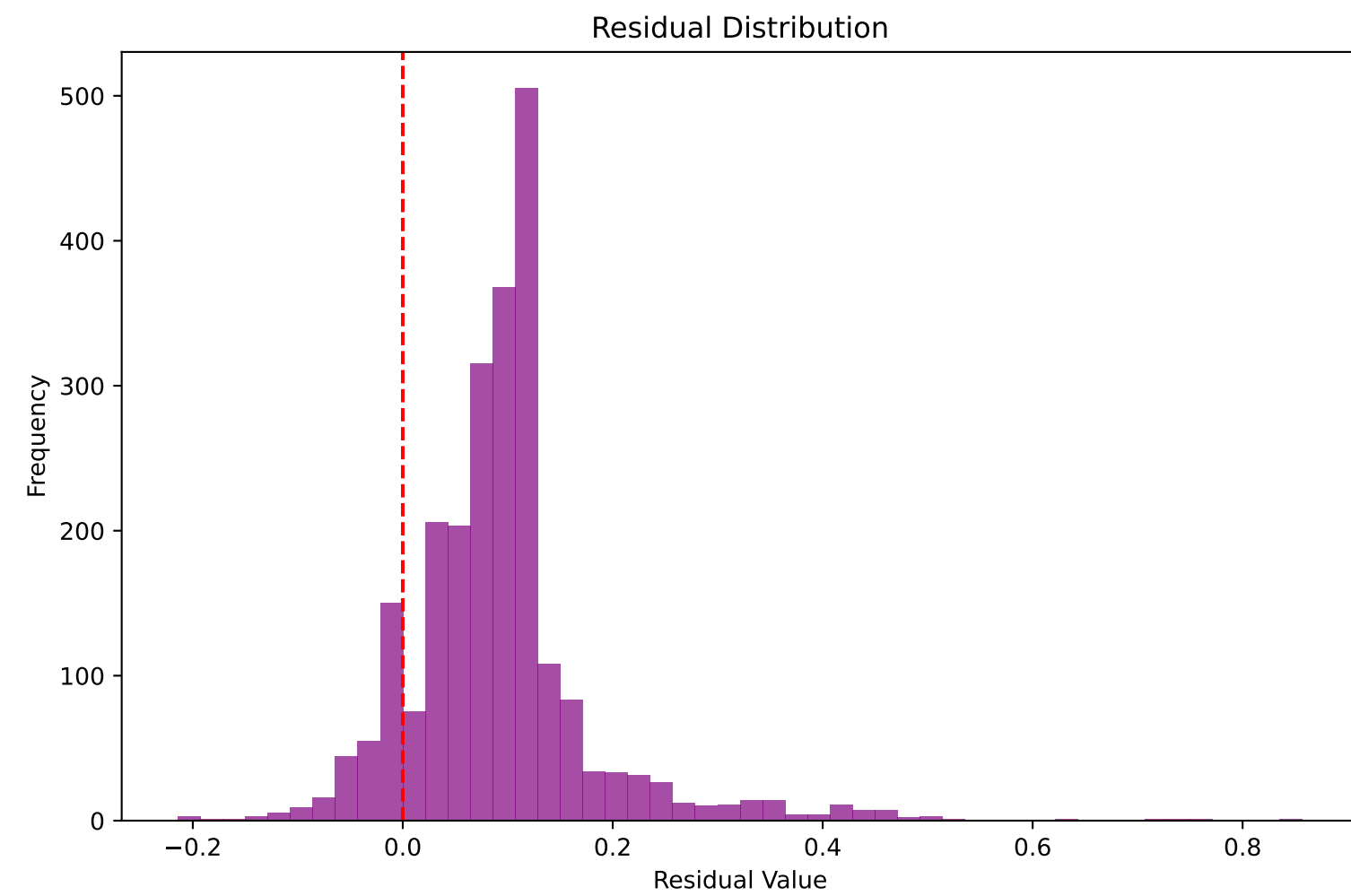
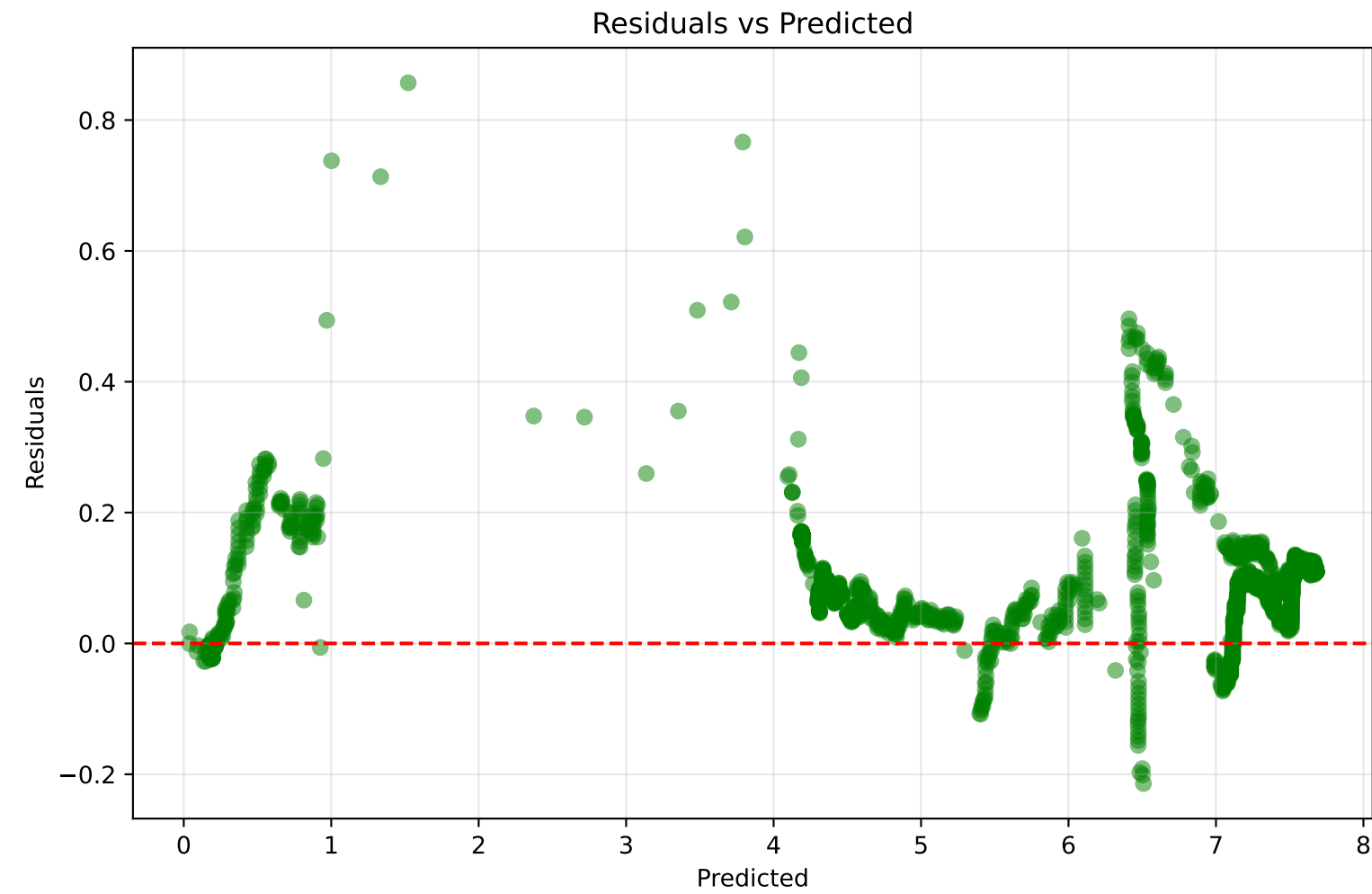
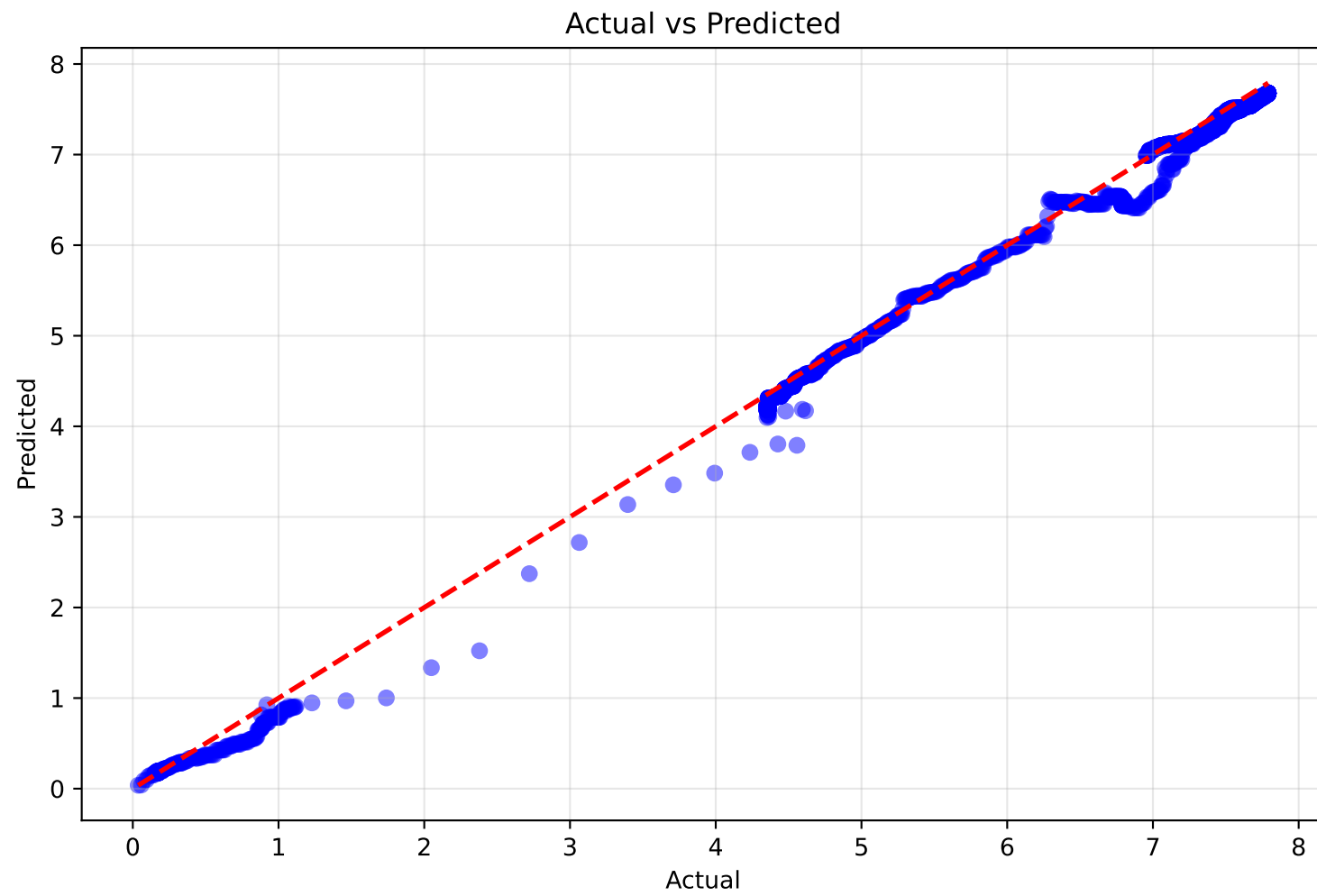
Residual Distribution



Zoomed View (First 200 points)

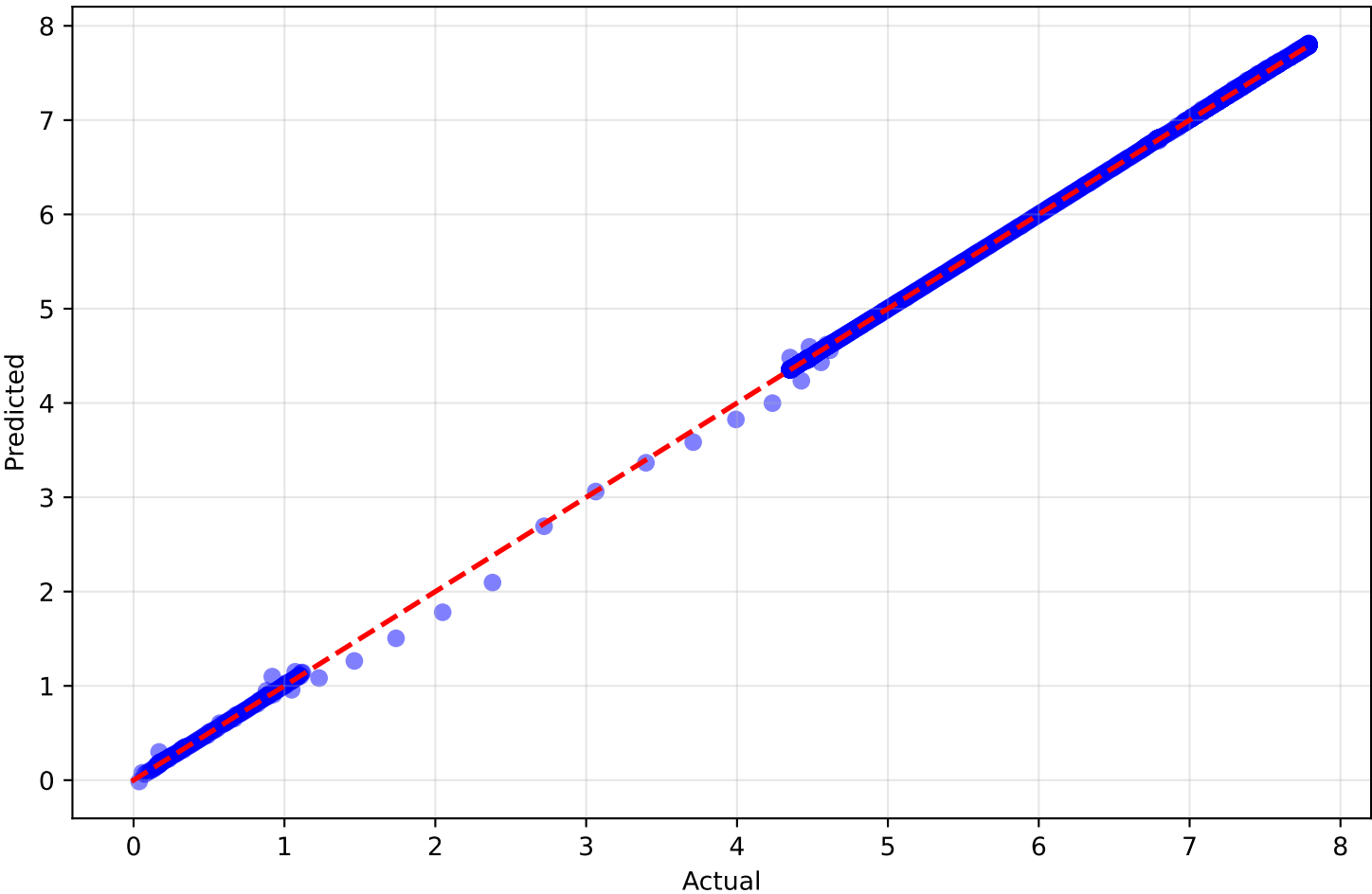


Detailed Analysis: XGBoost

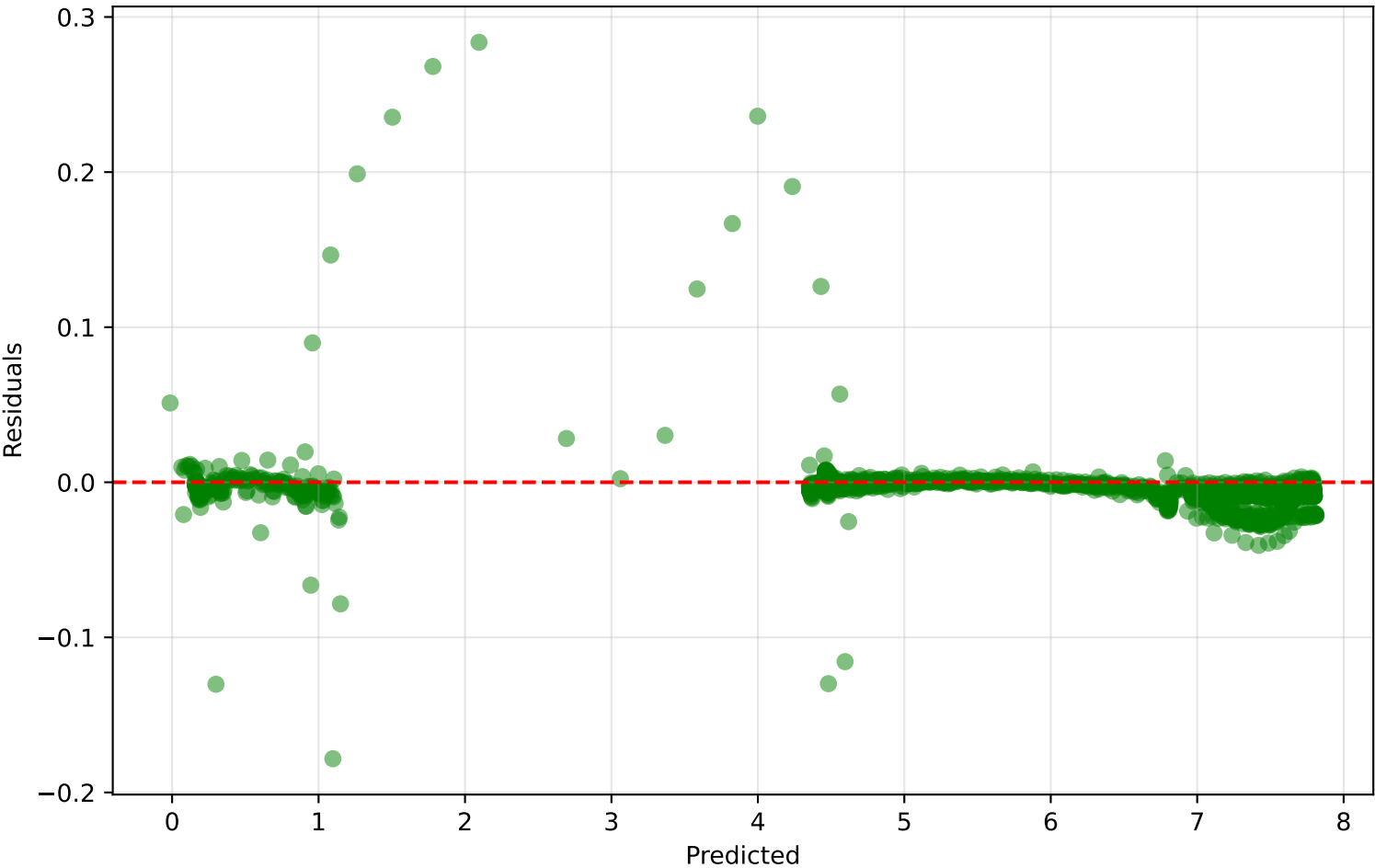


Detailed Analysis: Ridge

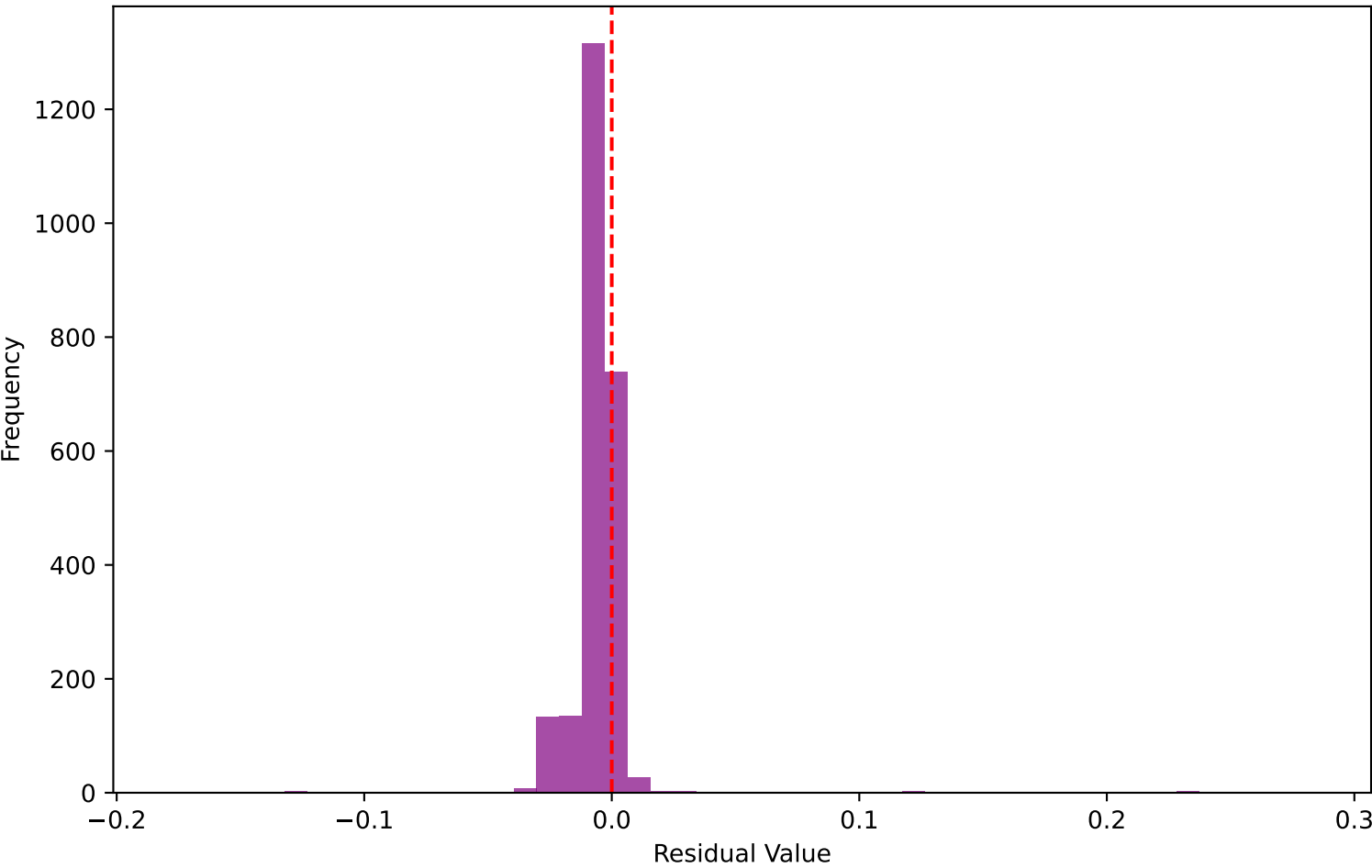
Actual vs Predicted



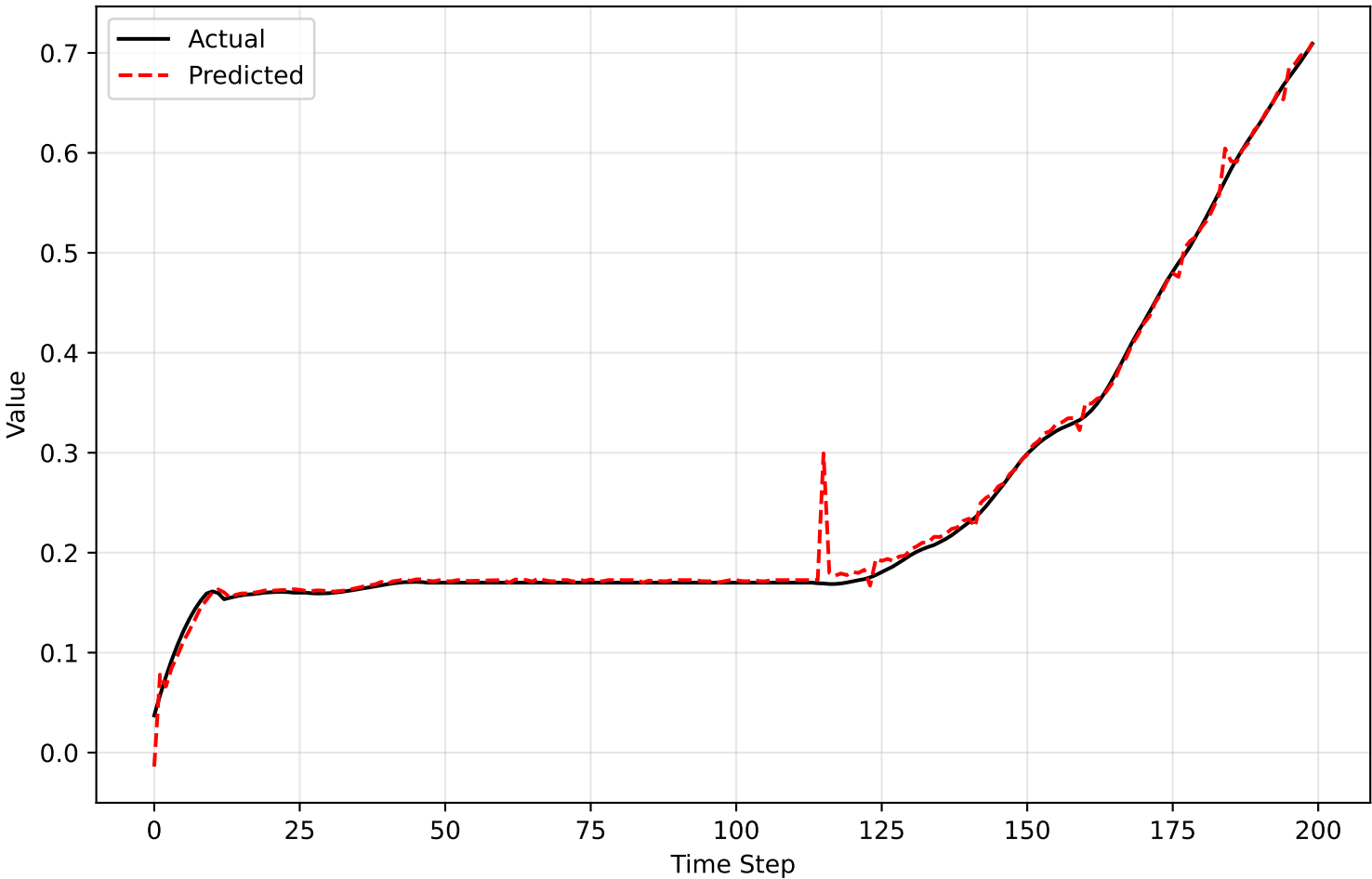
Residuals vs Predicted



Residual Distribution

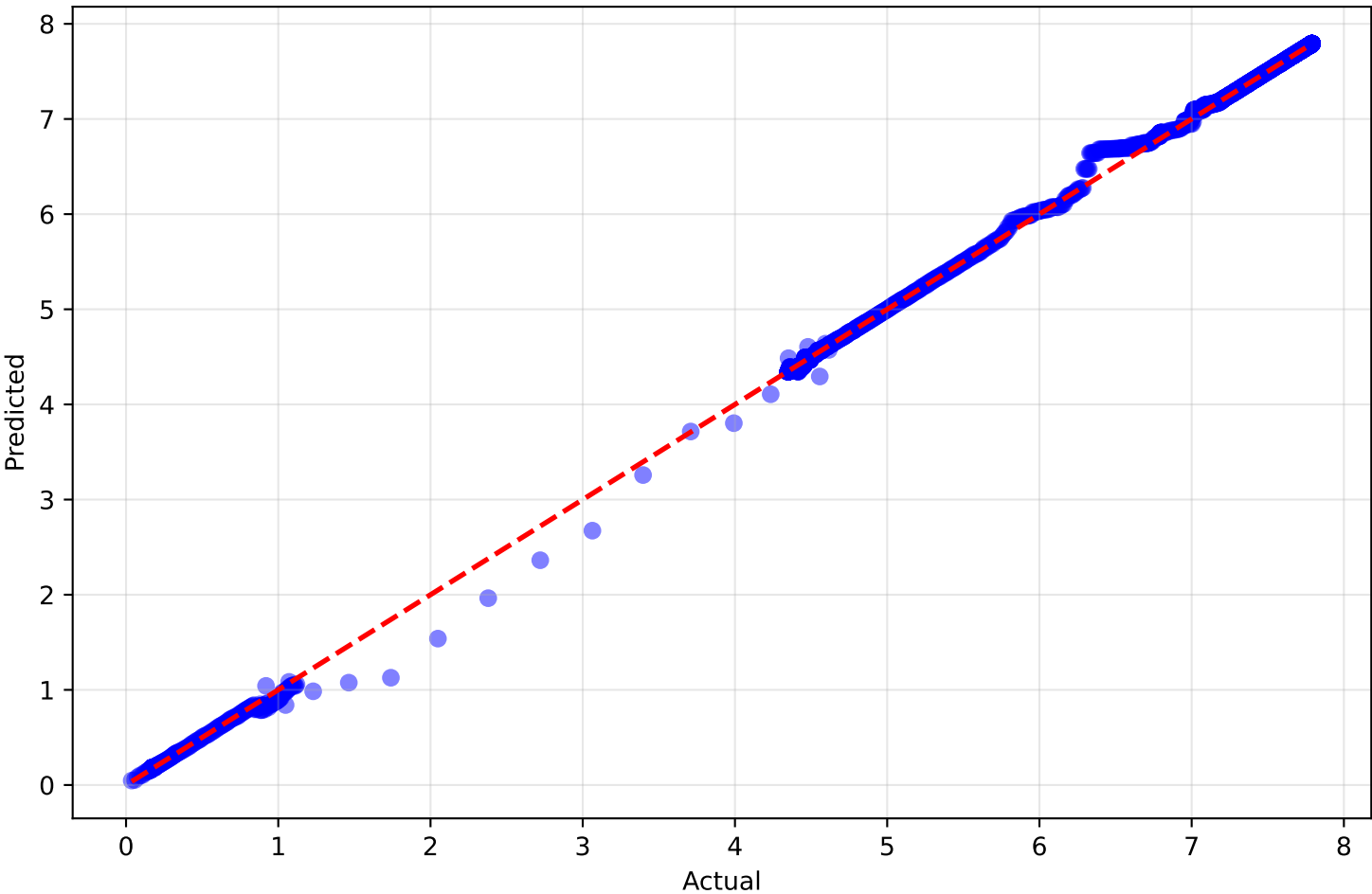


Zoomed View (First 200 points)

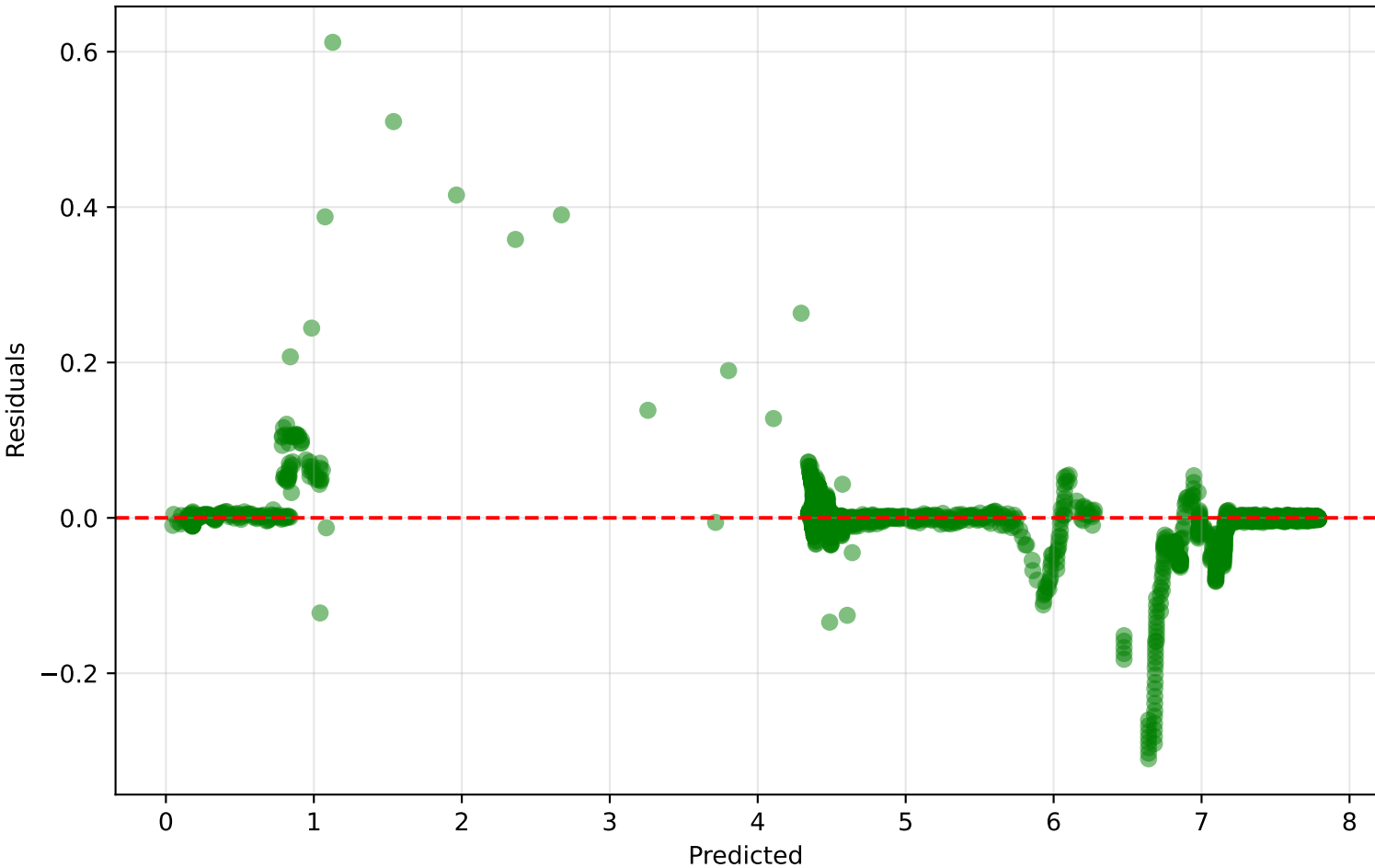


Detailed Analysis: Gradient Boosting

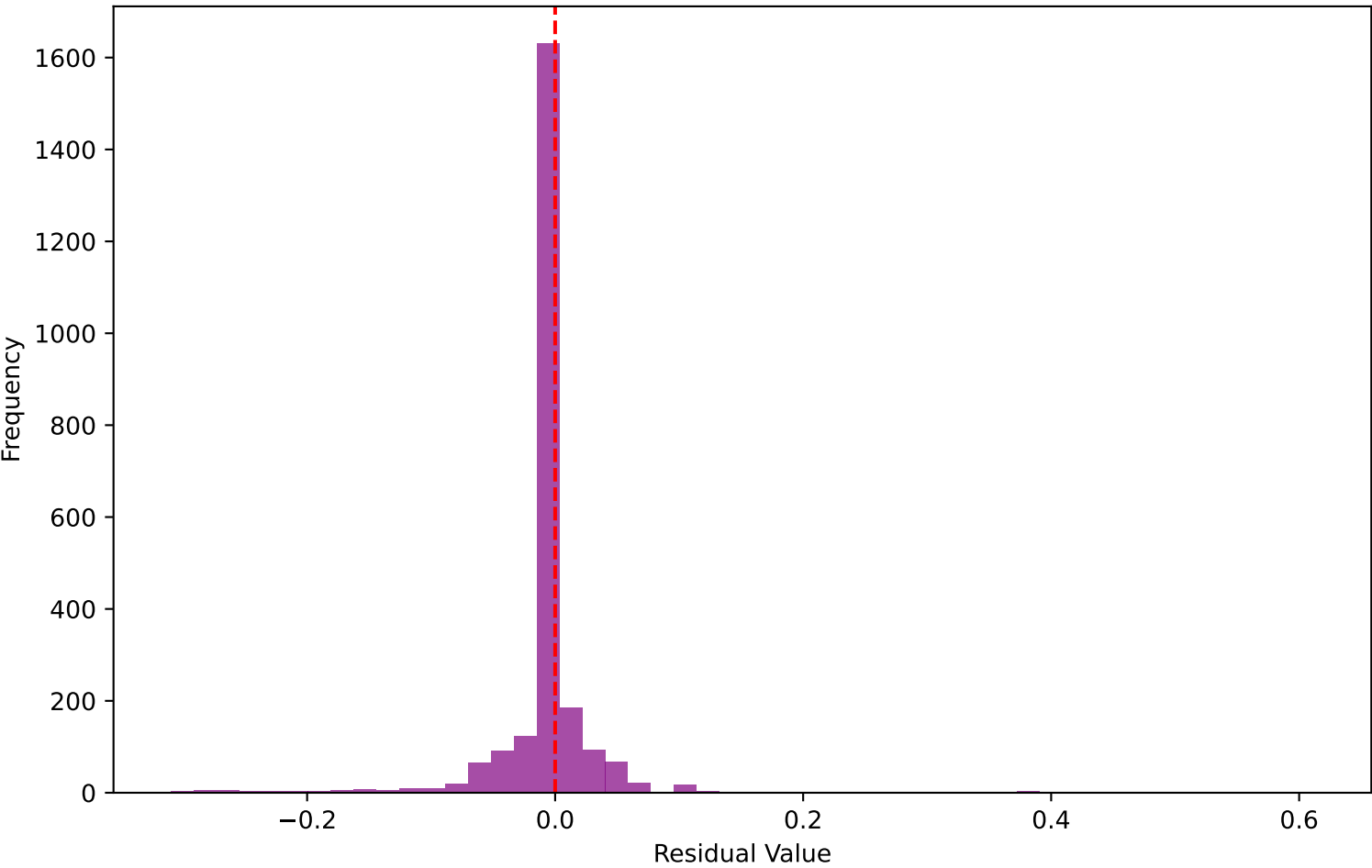
Actual vs Predicted



Residuals vs Predicted



Residual Distribution



Zoomed View (First 200 points)

